(6) Determine whether the vectors $(1,1,0,0),(0,-1,1,1),(2,1,-1,1)$ in $\mathbb{R}^4$ are linearly independent or not. [3]

$$\alpha(1,1,0,0)+\beta(0,-1,1,1)+\gamma(2,1,-1,1)=0$$

$$\alpha+2\gamma=0$$

$$\alpha-\beta+\gamma=0$$

$$\beta-\gamma=0$$

$$\beta+\gamma=0$$

$\Rightarrow \beta=0;\ \gamma=0;\ \alpha=0$

$\Rightarrow$ linearly independent

(6) (a) If $A \in \mathbb{R}^{2\times 1}$ and $B \in \mathbb{R}^{1\times 2}$, then prove that the matrix AB is not invertible. [2]

(b) Show that if a real matrix is orthogonal and triangular, then it must be diagonal. Determine the diagonal entries of such a matrix. [4]

(a) $A=\begin{bmatrix} a_1 \\ b_1 \end{bmatrix}$ $B=\begin{bmatrix} a_2 & b_2 \end{bmatrix}$ $AB=\begin{bmatrix} a_1a_2 & a_1b_2 \\ b_1a_2 & b_1b_2 \end{bmatrix}$

$|AB|=0 \Rightarrow$ non invertable

(8) Let $A=\begin{bmatrix} 1 & 1 & 2 & 0 \\ 3 & 5 & 7 & 1 \\ 1 & 5 & 4 & 5 \\ 2 & 8 & 7 & 9 \end{bmatrix}$.

(a) Find the RREF of A.

(b) Find the rank of A. $\rightarrow 3$

(c) Express one of the columns as a linear combination of the other columns. [6]

$\hookrightarrow C_3=3/2C_1+1/2C_2$

$$\begin{bmatrix} 1 & 1 & 2 & 0 \\ 3 & 5 & 7 & 1 \\ 1 & 5 & 4 & 5 \\ 2 & 8 & 7 & 9 \end{bmatrix} \xrightarrow[E_{-1}[3,1]]{E_{-3}[2,1]} \begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & 2 & 1 & 1 \\ 0 & 4 & 2 & 5 \\ 0 & 6 & 3 & 9 \end{bmatrix}$$

$E_{1/2}[1,2]$

$$\begin{bmatrix} 1 & 0 & 3/2 & -1/2 \\ 0 & 2 & 1 & 1 \\ 0 & 4 & 2 & 5 \\ 0 & 6 & 3 & 9 \end{bmatrix} \xrightarrow{E_{1/2}[2]} \begin{bmatrix} 1 & 0 & 3/2 & -1/2 \\ 0 & 1 & 1/2 & 1/2 \\ 0 & 4 & 2 & 5 \\ 0 & 6 & 3 & 9 \end{bmatrix}$$

$E_{-4}[3,2]$ $E_{-6}[4,2]$

$$\begin{bmatrix} 1 & 0 & 3/2 & -1/2 \\ 0 & 1 & 1/2 & 1/2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 6 \end{bmatrix} \xrightarrow[E_{-1/6}[2,3]]{E_{1/6}[4,3]} \begin{bmatrix} 1 & 0 & 3/2 & 0 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$E_{-2}[4,3]$

$C_3 = 3/2\,C_1 + 1/2\,C_2$

$E_{1/3}[3]$

$$\begin{bmatrix} \boxed{1} & 0 & 3/2 & 0 \\ 0 & \boxed{1} & 1/2 & 0 \\ 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

5. For what value of a and b, the rank of matrix A is 2.

$$A = \begin{bmatrix} 1 & -2 & 3 & 1 \\ 2 & 1 & -1 & 2 \\ 6 & -2 & a & b \end{bmatrix}$$

[3]

$$A \xrightarrow[E_{-6}[3,1]]{E_{-2}[2,1]} \begin{bmatrix} 1 & -2 & 3 & 1 \\ 0 & 5 & -7 & 0 \\ 0 & 10 & a-18 & b-6 \end{bmatrix}$$

$$\downarrow E_{1/5}[1,3]$$

$$\begin{bmatrix} 1 & 0 & 3+\frac{a-18}{5} & 1+\frac{b-6}{5} \\ 0 & 5 & -7 & 0 \\ 0 & 10 & a-18 & b-6 \end{bmatrix}$$

$$\downarrow E_{-2}[3,2] \; , \; E_{1/5}[2]$$

$$a=4, b=6 \Leftarrow \begin{bmatrix} 1 & 0 & 3+\frac{a-18}{5} & 1+\frac{b-6}{5} \\ 0 & 1 & -7/5 & 0 \\ 0 & 0 & a-4 & b-6 \end{bmatrix}$$

6. Suppose that the vectors $\{u, v, w\}$ are linearly independent. Show that the vectors $\{u+v, u-v, u-2v+w\}$ are also linearly independent. [3]

$$\alpha(u+v)+\beta(u-v)+\gamma(u-2v+w)=0$$

$$\Rightarrow u(\alpha+\beta+\gamma)+v(\alpha-\beta-2\gamma)+w(\gamma)=0$$

$$pu+qv+\gamma w=0 \Rightarrow p=0, q=0, \gamma=0$$

$$\Rightarrow \alpha+\beta+\gamma=0 \quad \alpha-\beta-2\gamma=0, \gamma=0$$

$$\Rightarrow \boxed{\alpha=\beta=\gamma=0}$$

7. Find the row reduced echelon form (RREF) of the matrix

$$A = \begin{bmatrix} 3 & -3 & -2 \\ -2 & 1 & 1 \\ -2 & 1 & -2 \\ -1 & 1 & 2 \end{bmatrix}$$

[4]

$$\begin{bmatrix} 3 & -3 & -2 \\ -2 & 1 & 1 \\ -2 & 1 & -2 \\ -1 & 1 & 2 \end{bmatrix} \xrightarrow[E_{2/3}[2,1]\ E_{2/3}[3,1]\ E_{1/3}[4,1]\ E_{1/3}[1]]{} \begin{bmatrix} 1 & -1 & -2/3 \\ 0 & -1 & -1/3 \\ 0 & -1 & -10/3 \\ 0 & 0 & 4/3 \end{bmatrix}$$

$$\downarrow \quad E_{-1}[1,2]\ E_{-1}[2]\ E_{-1}[3,2]$$

$$\begin{bmatrix} 1 & 0 & -1/3 \\ 0 & 1 & 1/3 \\ 0 & 0 & -3 \\ 0 & 0 & 4/3 \end{bmatrix} \xrightarrow[E_{4/9}[4,3]\ E_{-1/3}[3]]{E_{-1/9}[1,3]\ E_{1/9}[2,3]} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

(10) Consider the system of linear equations

$$5x + 3y + 8z = 5$$
$$3x + 2y + 5z = 6$$
$$px + 4y + 4z = q.$$

Find conditions on p and q so that the given system of linear equations has

(a) no solution; (b) unique solution; (c) infinitely many solutions. [7]

not possible $p \neq 0$ not possible

$$\left[\begin{array}{ccc|c} 5 & 3 & 8 & 5 \\ 3 & 2 & 5 & 6 \\ p & 4 & 4 & q \end{array}\right] \xrightarrow[E_{-p/5}[3,1]\ E_{1/5}[1]]{E_{-3/5}[2,1]} \left[\begin{array}{cccc} 1 & 3/5 & 8/5 & 1 \\ 0 & 1/5 & 1/5 & 3 \\ 0 & \frac{20-3p}{5} & \frac{20-8p}{5} & q-p \end{array}\right]$$

$E_5[2]$ ↓ $E_{-3}[1,2]$ $E_{3p-20}[3,2]$

$$\left[\begin{array}{cccc} 1 & 0 & 1 & -8 \\ 0 & 1 & 1 & 15 \\ 0 & 0 & -p & q+8p-60 \end{array}\right] \xrightarrow[E_{1/p}[2,3]\ E_{-1/p}[3]]{E_{1/p}[1,3]} \left[\begin{array}{cccc} 1 & 0 & 0 & \frac{q-60}{p} \\ 0 & 1 & 0 & 23+\frac{q-60}{p} \\ 0 & 0 & 1 & \frac{q+8p-60}{-p} \end{array}\right]$$

$(p \neq 0)$

↓ 1 soln

↓

$p=0$
$q=60$ } ∞

$p=0$
$q \neq 60 \rightarrow$ no

(12) Using the Cayley-Hamilton theorem, find the inverse of the matrix [4]

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 1 & 2 \\ -1 & 3 & 4 \end{bmatrix}.$$

$$\left[\begin{array}{ccc|ccc} 3 & -1 & 1 & 1 & 0 & 0 \\ -1 & 1 & 2 & 0 & 1 & 0 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{array}\right]$$

$$\begin{bmatrix} 1 & -1/3 & 1/3 & 1/3 & 0 & 0 \\ 0 & 2/3 & 7/3 & 1/3 & 1 & 0 \\ 0 & 8/3 & 13/3 & 1/3 & 0 & 1 \end{bmatrix}$$

$$\begin{matrix} 1 & 0 & 3/2 & 1/2 & 1/2 & 0 \\ 0 & 1 & 7/2 & 1/2 & 3/2 & 0 \\ 0 & 0 & -5 & -1 & -4 & 1 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 1/5 & -7/10 & 3/10 \\ 0 & 1 & 0 & -1/5 & 13/10 & 7/10 \\ 0 & 0 & 1 & 1/5 & 4/5 & -1/5 \end{bmatrix}$$

PART - B

8. Let $v_1 = (1,3,4)$, $v_2 = (-4,2,-6)$, $v_3 = (-3,-2,c)$. Find the value of c such that v_1, v_2, v_3 are linearly dependent. Using the obtained value of c, find α, β, γ such that $\alpha v_1 + \beta v_2 + \gamma v_3 = 0$. [5]

$$\theta_3 = p\theta_1 + q\theta_2$$

$$(-3,-2,c) = (p-4q, 3p+2q, 4p-6q)$$

$p - 4q = -3$

$3p + 2q = -2 \implies c = 4p - 6q = -4 - 3 = -7$

$p = -1, \quad q = 1/2$

$\alpha = -1, \beta = 1/2, \gamma = -1$

v_1, v_2, v_3. [4]

10. Using Gauss-Jordan elimination method, determine the values of $k \in \mathbb{R}$ so that the system of linear equations

$$\begin{aligned} x + y + z &= 1, \\ 2x + 3y + kz &= 3, \\ x + ky + 3z &= 3. \end{aligned}$$

has (a) no solution (b) infinitely many solutions (c) exactly one solution. Also, find solutions of the given system, if it exists.

[9]

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & k & 3 \\ 1 & k & 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & k-2 & 1 \\ 0 & k-1 & 2 & 2 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & k/2 \\ 0 & 0 & 1 & 1/k \end{bmatrix} \leftarrow \begin{bmatrix} 1 & 0 & 3-k & 0 \\ 0 & 1 & k-2 & 1 \\ 0 & 0 & 3k-k^2 & 3-k \end{bmatrix}$$

$k \neq 3$

$k \neq 0$

$k = 0 \rightarrow$ no solution

$k = 3 \rightarrow \infty$ solution

$k \neq 0, 3 \rightarrow 1$ solution